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SUPERSEDING
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23 March 2018

DETAIL SPECIFICATION

ALUMINUM POWDER, FLAKED, GRAINED, AND ATOMIZED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers aluminum powder for use in pyrotechnics, incendiaries, propellants, and explosives.

1.2 Classification. The aluminum powder types, grades, and classes should be as listed in Table I (see 6.1.1 and 6.3).

TABLE I. Classification.

Type	Grade	Class
Type I - flaked	A	1
	B	2
		2A
		3
Type II – grained or atomized	C	4
		4A
		5
		5A
		6
Type III - atomized	C	6
		7
		8
	G	9
	H	10

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-313 - Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-14067 - Powders, Metal (for use in Ammunition)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1234 - Pyrotechnics: Sampling, Inspection and Testing
MIL-STD-1916 - DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government standards and publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B214 - Standard Test Method for Sieve Analysis of Metal Powders (DoD adopted).
ASTM B215 - Standard Practices for Sampling Metal Powders
ASTM B329 - Standard Test Method for Apparent Density of Metal Powders and Compounds Using the Scott Volumeter (DoD adopted).

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ASTM B330	-	Standard Test Methods for Estimating Average Particle Size of Metal Powders and Related Compounds Using Air Permeability
ASTM B417	-	Standard Test Methods for Apparent Density of Non-Free-Flowing Metal Powders Using the Carney Funnel.
ASTM D480	-	Standard Test Methods for Sampling and Testing of Flaked Aluminum Powders and Pastes (DoD adopted).
ASTM E11	-	Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves (DoD adopted).
ASTM E300	-	Standard Practice for Sampling Industrial Chemicals.
ASTM E3061	-	Standard Test Method for Analysis of Aluminum and Aluminum Alloys by Inductively Coupled Plasma Atomic Emission Spectrometry (Performance Based Method)

(Applications for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959. Electronic copies of ASTM documents are available from https://www.astm.org/).

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.3), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with 4.3.

3.2 Material. Aluminum powder shall be manufactured from aluminum metal of such purity that the product meets the requirements of Table II. Historic references to Grades D, E, and F shall mean Grade C. Historic references to MIL-DTL-14067 Type II shall mean Grade G, and MIL-DTL-14067 Type V shall mean Grade H.

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TABLE II. Chemical characteristics.

Requirement, weight %	Grade A	Grade B	Grade C	Grade G	Grade H
Al, min.	98.0	98.0	99.0	98.75	99.00
Cu, max.	0.50	0.50	0.50	-	-
Fe, max.	0.50	0.50	0.50	0.55	0.20
Si, max.	0.50	0.50	0.50	0.25	-
Mg, max.	0.10	0.10	0.10	-	-
Zn, max.	0.25	0.25	0.25	0.3	-
Nonvolatile matter, min.	99.0	99.5	-	99.9	99.9
Easily extracted fatty and oil matter, max.	1.0	1.5	-	0.2	0.2

3.3 Form.

3.3.1 Type I. Type I aluminum powder shall be in the form of irregular flat flakes when examined as specified in 4.5.10.

3.3.2 Type II. Type II aluminum powder shall consist of granular particles of nodular or spheroid form flakes when examined as specified in 4.5.10.

3.3.3 Type III. Type III aluminum powder shall consist of granular particles of spheroid form flakes when examined as specified in 4.5.10.

3.4 Particle size distribution. Types I, II, and III aluminum powder shall conform to the requirements of Table III when tested as specified in 4.5.11. The powder shall pass through the required sieves readily without balling or the particles clinging together.

Table III. Particle distribution.

Classes	Density(g/cm^3)	Size range mesh[1] (µm)	Weight %
1	0.30 max.	+325 mesh (>45) -325 mesh (<45)	1.0 max. 99.0 min.
2	No determination	+325 mesh (>45) -325 mesh (<45)	6.0 max. 94.0 min.
2A	No determination	+100 mesh (>150) -100 to +200 mesh (150 - 75) -200 mesh (<75)	0.2 max. 20.0 max. 80.0 min.
3	0.50 max.	+100 mesh (>150) -100 to +200 mesh (150 - 75) -200 to +325 mesh (75 - 45) -325 mesh (<45)	0.2 max. 15.0 max. 15.0 max. 80.0 min.
4	0.90 min.	+50 mesh (>300) -50 to +100 mesh (50 - 150) -100 to +200 mesh (150 - 75) -200 mesh (<75)	0.0 3.0 max. 3.0 - 20.0 80.0 - 97.0

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Table III. Particle distribution (continued).

Classes	Density(g/cm^3)	Size range mesh[1] (µm)	Weight %
4A	0.90 min.	+20 mesh (>850) -20 to +100 mesh (850 - 150) -100 to +200 mesh (150 - 75) -200 mesh (<75)	2.0 max. 30.0 max. 65.0 max. 30.0 - 60.0
5	0.90 min.	+12 mesh (>1700) -12 to +30 mesh (1700 - 600) -30 mesh (<600)	0.0 13.0 - 26.0 74.0 - 87.0
5A	0.90 min.	+30 mesh (>600) -30 to +50 mesh (600 - 300) -50 to +140 mesh (300 - 106) -140 mesh (<106)	13.0 - 26.0 35.0 - 45.0 30.0 - 46.0 3.0 max.
6	0.95 to 1.20	+100 mesh (>150) -100 to +200 mesh (150 - 75) -200 to +325 mesh (75 - 45) -325 mesh (<45)	2.0 max. 20.0 max. 10.0 - 35.0 65.0 - 90.0
7	0.95 min.	+40 mesh (>425) -325 mesh (<45)	0.5 max. 25.0 - 50.0
8	0.95 min.	+12 mesh (>1700) -325 mesh (<45)	0.5 max. 35.0 max.
9	1.0 min.	-200 (< 75) -325 (< 45)	99.5 min. 60.0 min.
10	No determination	No determination	No determination

[1]A + sign means particles are larger than the nominal sieve size. A – sign means particles are smaller than the nominal sieve size.

3.5 Average particle size.

3.5.1 Particle size for Class 9. The average particle size for Type III Grade G aluminum powder shall be 22 +/- 8 microns when tested as specified in 4.5.12.

3.5.2 Particle size for Class 10. The average particle size for Type III Grade H aluminum powder shall be 25 +/- 5 microns when tested as specified in 4.5.12.

3.6 Apparent density. The apparent density in grams per cubic centimeter shall conform to the requirements of Table III when tested as specified in 4.5.13.

3.7 Process verification. All types of aluminum powder shall be manufactured by the process classified in Table I (see 6.4).

3.8 Workmanship of metal powder. The metal powder shall be uniform in quality and free from lumps, grit, visible impurities, foreign matter, or other defects that would render the metal powder unsuitable for intended use.

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4. VERIFICATION

TABLE IV. Requirements/verification cross reference matrix.

Method of verification 1 – Analysis 2 – Demonstration 3 – Examination 4 – Test						Classes of verification A – First article B – Conformance		
Section 3 Requirement	Description	Verification Methods				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article	X	X	X	X	X		4.2
3.1.2	Conformance	X	X	X	X		X	4.3
3.2	Aluminum	X		X	X	X	X	4.5.1.4 or 4.5.2
3.2	Copper	X		X	X	X	X	4.5.1 or 4.5.3
3.2	Iron	X		X	X	X	X	4.5.1 or 4.5.4
3.2	Silicon	X		X	X	X	X	4.5.5
3.2	Magnesium	X		X	X	X	X	4.5.1 or 4.5.6
3.2	Zinc	X		X	X	X	X	4.5.1 or 4.5.7
3.2	Nonvolatile matter	X		X	X	X	X	4.5.8
3.2	Fatty and oil matter	X		X	X	X	X	4.5.9
3.3	Form			X	X	X	X	4.5.10
3.4	Particle size distribution			X	X	X	X	4.5.11
3.5	Average particle size			X	X	X	X	4.5.12
3.6	Apparent density			X	X	X	X	4.5.13
3.7	Process verification		X	X		X	X	4.5.14
3.8	Workmanship			X		X	X	4.5.15

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see paragraph 4.2).
b. Conformance inspection (see paragraph 4.3).

4.2 First article inspection. When specified, a sample shall be subject to first article verification in accordance with Table V.

4.2.1 First article quantity. The first article sample shall be a minimum of 10 kilograms of metal powder from the first production lot produced by the contractor using the same production equipment, process procedures, and material supply sources as will be used in regular production.

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4.2.2 Inspections to be performed. The first article sample shall be subjected to inspection and tests specified in Table V and 4.5.

4.2.3 Rejection. If any sample fails to comply with any of the applicable requirements the first article quantity shall be rejected.

TABLE V. First article inspection.

Classification	Metal powder	Requirement paragraph	Inspection method reference
Critical	None defined		
Major			
101	Aluminum	3.2	4.5.1.4 or 4.5.2
102	Copper	3.2	4.5.1 or 4.5.3
103	Iron	3.2	4.5.1 or 4.5.4
104	Silicon	3.2	4.5.5
105	Magnesium	3.2	4.5.1 or 4.5.6
106	Zinc	3.2	4.5.1 or 4.5.7
107	Nonvolatile matter	3.2	4.5.8
108	Fatty and oil matter	3.2	4.5.9
109	Form	3.3	4.5.10
110	Particle size distribution	3.4	4.5.11
111	Average particle size	3.5	4.5.12
112	Apparent density	3.6	4.5.13
Minor			
201	Process verification	3.7	4.5.14
202	Workmanship	3.8	4.5.15

4.3 Conformance inspection.

4.3.1 Inspection lot formation. A lot shall consist of one or more batches (see 6.4) from no greater than one month of production of the aluminum powder of the same type, grade, and class offered for acceptance at one time. Lot formation shall be in accordance with the lot formation requirements of MIL-STD-1916. Each lot shall consist of that quantity of aluminum powder that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous.

4.3.2 Classification of characteristics. For the conformance inspection paragraph 4.3.3.1, the definitions of critical, major, and minor defects are provided in paragraph “Definitions” of MIL-STD-1916.

4.3.3 Inspection requirements by classification of characteristics.

a. The characteristics shown in 3.2 thru 3.8 when tested in accordance with 4.5 shall constitute minimum inspections to be performed.

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4.3.3.1	Metal Powder	Conformance criteria	Requirement paragraph	Inspection method reference
Critical	None defined			
Major				
101	Aluminum	4.4.1	3.2	4.5.1.4 or 4.5.2
102	Copper	4.4.1	3.2	4.5.1 or 4.5.3
103	Iron	4.4.1	3.2	4.5.1 or 4.5.4
104	Silicon	4.4.1	3.2	4.5.5
105	Magnesium	4.4.1	3.2	4.5.1 or 4.5.6
106	Zinc	4.4.1	3.2	4.5.1 or 4.5.7
107	Nonvolatile matter	4.4.1	3.2	4.5.8
108	Fatty and oil matter	4.4.1	3.2	4.5.9
109	Form	4.4.1	3.3	4.5.10
110	Particle size distribution	4.4.1	3.4	4.5.11
111	Average particle size	4.4.1	3.5	4.5.12
112	Apparent density	4.4.1	3.6	4.5.13
Minor				
201	Workmanship	4.4.1	3.8	4.5.15

4.4 Metal powder sampling.

4.4.1 Sampling of lot. Sample containers shall be obtained at random from each lot of aluminum powder in accordance with Table VI. When lots exceed 2,500 containers, the sample size shall be calculated using the following equation

$$n = 0.15\sqrt{N}$$

Where:

n = sample size
N = lot size

A specimen, minimum two kilograms, shall be obtained from each container in the sample using ASTM E300 or ASTM B215. Each specimen shall be tested as specified in 4.5. Failure of any test, by any sample, shall be cause for rejection of the lot represented.

TABLE VI. Sampling.

Containers in a Lot	Number of Samples
1	1
2 to 275	2
276 to 545	3
546 to 900	4
901 to 1345	5
1346 to 1875	6
1876 to 2500	7

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4.5 Test methods and procedures. Unless otherwise noted, distilled water and analytical grade reagents shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant.

4.5.1 Chemical composition (Inductively Coupled Plasma (ICP) method) (preferred method). For the determination of Cu, Fe, Mg, and Zn in Al grades A, B, and C. Zn and Fe in Grade G. Fe in Grade H.

4.5.1.1 Reagents.

a. HCl: Hydrochloric Acid 35-38% trace metal analysis grade
b. Hydrogen Peroxide: 30% ACS grade or higher
c. Water: Unless otherwise noted, references to water shall mean laboratory accepted deionized water.

4.5.1.2 Stock solutions. It is advisable to use commercially available stock solutions with certified concentrations traceable to primary standards (National Institute of Standards and Technology or international measurement standards). Stock solutions shall be replaced after one year or in accordance with manufacturer recommendations. Recommended stock concentrations as follows:

a. Aluminum ICP standard 20,000 ppm
b. Copper ICP standard 1,000 ppm
c. Iron ICP standard 1,000 ppm
d. Zinc ICP standard 500 ppm
e. Magnesium ICP standard 200 ppm

Alternately, stock solutions may be prepared in-house from high purity metals or their salts. The maximum shelf life is one year from date of initial preparation.

4.5.1.3 Procedure.

4.5.1.3.1 Standard preparation.

a. Mid Standard: Add stepwise to a 200 mL volumetric flask: 50 mL Al solution, 4 mL Fe solution, 4 mL Cu solution, 4 mL Zn solution, 4 mL Mg solution. Cap flask and mix contents for 5 minutes. Allow solution to reach ambient temperature and fill to 200 mL mark with water.

b. High Standard: Add stepwise to a 200 mL volumetric flask: 50 mL Al solution, 8 mL Fe solution, 8 mL Cu solution, 8 mL Zn solution, 8 mL Mg solution. Cap flask and mix contents for 5 minutes. Allow solution to reach ambient temperature and fill to 200 mL mark with water.

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c. Blank standard: Whether using in-house or commercially prepared standards, the blank shall be prepared so the resulting matrix closely matches that of the standards.

4.5.1.3.2 Sample preparation.

a. Weigh 0.5 gram of sample to the nearest 1 milligram and transfer into 300 mL beaker.
b. Add 10 mL of HCl and cover with watch glass and allow reaction to subside.
c. Add 1 mL hydrogen peroxide and allow reaction to subside.
d. Gently swirl beaker for 5 minutes to complete dissolution.
e. Rinse watch glass with water into beaker.
f. Allow to cool to ambient temperature.
g. Transfer contents of beaker into 100 mL volumetric flask. Rinse beaker with water into flask.
h. Allow solution to reach ambient temperature and fill to 100 mL mark with water.

4.5.1.3.3 Preparation of calibration curve.

a. Prepare standards as described in 4.5.1.3.1.
   1. Mid Standard shall contain: 20.0 ppm each of Cu and Fe; 10.0 ppm Zn; 4.0 ppm Mg.
   2. High Standard shall contain: 30.0 ppm each of Cu and Fe; 15.0 ppm Zn; 6.0 ppm Mg.
   3. Blank shall contain: 5,000 µg/mL Al.

b. Emission intensities shall be measured using Radial view at the following wavelengths:
   1. Cu: 324.754 nm
   2. Fe: 259.940 nm
   3. Zn: 213.856 nm
   4. Mg: 279.553 nm

c. Aspirate standard solutions into ICP spectrometer in order of increasing concentration.

d. Take emission measurements for each element and generate a calibration curve utilizing instrument software. Repeat calibration if the correlation determination (R_2) of any element is < 0.9999. If calibration fails this criteria, repeat the analysis. If criteria is still not met, re-prepare the standards and perform the calibration again.

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4.5.1.3.4 Sample analysis.

a. Aspirate sample solution into ICP spectrometer and record emission measurement for Cu, Fe, Zn, and Mg.
b. Software generated concentrations for the elements shall be individually inserted into the formula below as A_1.
c. Aspirate blank solution containing 5,000 µg/mL Al and insert into formula below as A_2.

$$X = 100 \cdot (A_1 - A_2)/M$$

Where:
X = content of the element of interest in sample [% (mass fraction)]
A_1 = amount of the element of interest detected in the sample (g)
A_2 = amount of element of interest detected in the blank test solution (g)
M = amount of sample weighed out (g)

4.5.1.4 Aluminum content. Calculate the aluminum content by subtracting the results of 4.5.1.3.4 (or alternate method when used), 4.5.5 (Si), and 4.5.9 (fatty and oily matter) from 100.

$$\text{Al content (\% mass fraction)} = 100 - \sum X_n$$

Where:
X = content of the element of interest in sample [% (mass fraction)]
n = element of interest

4.5.2 Free metallic aluminum for Grades G and H (eudiometer method) (alternate method). The free metallic aluminum shall be determined as specified in method 412.1 of Standard MIL-STD-1234. Except that the reaction flask shall contain 100 milliliters of 10 percent sodium hydroxide solution saturated with hydrogen. At standard conditions a 0.33 to 0.34 grams sample shall be used and calculated as follows:

$$\text{Percent free metallic aluminum} = \frac{0.0288 \cdot V \cdot (P_1 - P_2)}{W \cdot (273 + T)} - (0.275 \cdot A + 1.28 \cdot B)$$

Where:
T = temperature of water jacket in C°.
V = volume of gas in buret, milliliter.
P_1 = barometric pressure, in millimeter of mercury.
P_2 = vapor pressure of water at T, in millimeter, of mercury.
W = weight of sample, grams.
A = percent zinc (see 4.5.1 or 4.5.7).
B = percent silicon (see 4.5.5).

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4.5.3 Copper (alternate method). The copper shall be determined in accordance with ASTM E3061.

4.5.4 Total Fe (alternate method). The total iron shall be determined in accordance with ASTM E3061.

4.5.5 Silicon as Si. Verify the silicon content by photometric examination (see 6.7).

4.5.5.1 Reagents.

a. Aluminum nitrate solution ($Al(NO_3)_3$) – Transfer 1.0 g of aluminum (99.9% purity or higher), low-silicon, to a 250-mL nickel beaker. Add 100 mL of NaOH solution and cover. Allow to react without applying heat until the reaction subsides, and then warm gently to complete the dissolution. Cool, and transfer to a 400 mL beaker containing 125 mL of HNO_3 (1+1). Cover and warm gently until the salts dissolve and the solution becomes clear. Cool, transfer to a 250 mL volumetric flask, dilute with distilled water to volume, and mix by inverting flask five (5) or more times.
b. Ammonium molybdate solution (100 g/L) – Dissolve 100 g of ammonium molybdate tetrahydrate ($(NH_4)_6Mo_7O_{24}•4H_2O$) in water and dilute to 1 L. Store in a plastic bottle. Before using, filter through a plastic funnel and Whatman 41 filter paper or equivalent. If a precipitate forms on standing, the solution should be discarded.
c. Silicon, standard solution (1 mL = 0.05 mg Si) – Fuse 0.1070 g of anhydrous silicon dioxide (SiO_2) with 1.0 g of sodium carbonate (Na_2CO_3) in a covered platinum crucible. Cool, dissolve the melt completely in water in a plastic beaker, cover, and heat on a steam bath for ½ to 1 hour. Cool, transfer to a 1 L volumetric flask, dilute to volume, and mix by inverting flask five (5) or more times. Store the solution in a plastic bottle. If anhydrous SiO_2 is not available, the weight should be adjusted according to the actual silicon content of the SiO_2 used, as determined by gravimetric analysis.
d. Sodium hydroxide solution (NaOH) (300 g/L) – Dissolve 300 g of sodium hydroxide (NaOH) in about 600 mL of water, using a nickel or stainless steel beaker. Cool and dilute to 1 L. Store in a plastic bottle.
e. Nitric acid solution (HNO_3 (1+1)) – Thoroughly mix one part reagent-grade hydrogen nitrate (HNO_3(aqueous) (approximately 70%)) with one part deionized water.

4.5.5.2 Procedure.

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4.5.5.2.1 Preparation of calibration curve.

a. Calibration solutions – Transfer 12.5 mL of $Al(NO_3)_3$ (see 4.5.5.1.a) solution and 1.0 mL of HNO_3 (1+1) to seven 100 mL beakers. Transfer 1.0, 2.0, 4.0, 8.0, 12.0, 16.0, and 20.0 mL of silicon solution (1 mL = 0.05 mg Si) to each of the 100 mL beakers. Dilute to 80 mL and mix. Proceed as directed in 4.5.5.2.1.c.
b. Reference solution – Transfer 12.5 mL of $Al(NO_3)_3$ solution and 1 mL of HNO_3 (1+1) to a 100 mL beaker, dilute to 80 mL and mix. Proceed as directed in 4.5.5.2.1.c.
c. Color development – Using a pH meter, check the pH of the solution. If the pH is not within 1.1 to 1.3, adjust it to this range using HNO_3 (1+1) or NaOH solution as required. Add 10 mL of ammonium molybdate solution, transfer the solution to a 100 mL volumetric flask, dilute to volume, and mix. Let stand at least 5 minutes but no longer than 20 minutes before taking the photometric reading.
d. Photometry – Transfer a suitable portion of the reference solution to an absorption cell with a 2-cm light path and adjust the photometer to the initial setting, using a light band centered at approximately 400+. While maintaining this adjustment, take the photometric readings of the calibration solutions.
e. Calibration curve – Plot the photometric readings of the calibration solutions against milligrams of silicon per 100 mL of solution.

4.5.5.2.2 Sample analysis.

a. Test solution
   i. Transfer 0.10 g of the sample, weighted to the nearest 0.1 mg, to a 50 mL nickel crucible, add 10.00 mL of NaOH solution, and cover. Allow to react without applying heat until the reaction subsides, wash down the cover and sides of the crucible with a minimum amount of water, and boil gently until the reaction is complete and the silicon completely oxidized (see 6.7.1). Cool, dilute to approximately 30 mL with water, and transfer quantitatively to a 150 mL beaker containing 12.5 mL of HNO_3 (1+1). Cover and warm gently until the solution becomes clear (see 6.7.2-3). Cool, transfer to a 100 mL volumetric flask, dilute to volume, and mix by inverting flask five (5) or more times until solution is visually homogenous.
   ii. Transfer a 50 mL aliquot to a 100 mL beaker, add 1.0 mL of HNO_3 (1+1), dilute to approximately 80 mL, and mix by inverting flask five (5) or more times until solution is visually homogenous.
b. Reagent blank
   i. Transfer 10.0 mL of NaOH solution to a 100 mL volumetric flask containing 11.5 mL of HNO_3 (1+1). Dilute to approximately 50 mL, cool, dilute to volume, and mix. Transfer a 50 mL aliquot to a 100 mL beaker, add 1 mL of HNO_3 (1+1), dilute to 80 mL, and mix by inverting flask five (5) or more times until solution is visually homogenous.

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ii. To the remaining portion of the reagent blank add 1 mL of HNO_3 (1+1), dilute to volume, and mix. This portion is used as the reagent blank for the reference solution in 4.5.5.2.2.d.

c. Color development – Proceed as directed in 4.5.5.2.1.c.

d. Reference solution – To the remaining portion of the test solution add 1 ml of HNO_3 (1+1), dilute to volume, and mix. Take the photometric reading of the reference solution as directed in 4.5.5.2.1.d using the reagent blank obtained in 4.5.5.2.2.b.ii for the initial setting of the photometer.

4.5.5.2.3 Calculation. Convert the photometric readings of the test and reference solutions to milligrams of silicon by means of the calibration curve. Calculate the percentage of silicon as follows:

$$Silicon, \% = \frac{(A - B)}{(C \times 10)}$$

Where:

A = silicon found in 100 mL of the final solution, mg
B = reference solution correction, expressed as milligrams of silicon
C = sample represented in 100 mL of the final solution, g

4.5.6 Magnesium (alternate method).

4.5.6.1 Mg for Grades A, B, and C. The free metallic magnesium shall be determined in accordance with ASTM E3061.

4.5.7 Zinc by atomic absorption method (alternate method). The zinc content shall be determined in accordance with ASTM E3061.

4.5.8 Non-volatile matter. Approximately 5 grams of the sample shall be weighed into a tared dish and heated in an oven for 3 hours at 105ºC. Remove from oven and place into a desiccator to cool. Reweigh sample and the loss in weight shall be calculated as percent non-volatile material at 105ºC.

$$\text{Percent non-volatiles} = \left(1 - \frac{\text{A}}{\text{B}}\right) \times 100$$

Where:

A = loss in weight, g
B = weight of sample, g

4.5.9 Fatty and oil matter.

4.5.9.1 Fatty and oil matter (Grades A, B, and C). Determine the fatty and oil matter content in accordance with ASTM D480.

4.5.9.2 Fatty and oil matter (Grades G and H).

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a. A weighed portion of approximately 20 grams of the sample shall be extracted with diethyl ether or hexane (see 6.6) in a Soxhlet or similar extractor using a tared flask.
b. The ether or hexane shall be evaporated by steam bath when extraction is completed in the flask.
c. The contents shall be dried at 90°C, to constant weight, cooled in a desiccator, and weighed. A blank shall be run at the same time.
d. The weight of the residue shall be calculated to percent oil and grease, as follows:

$$\text{Percent oil and grease} = \frac{100 \cdot (A-B)}{W}$$

Where:
A = weight of ether or hexane extract.
B = weight of blank.
W = weight of sample.

4.5.10 Form. Place a few grams of the sample on a glass slide and examine the material under a 20 to 100 power microscope such that a representative sample size is visible. Visually examine the sample for conformance.

4.5.11 Particle size distribution.

4.5.11.1 Particle size distribution (Class 1). The particle size distribution of Class 1 material shall be determined in accordance with ASTM D480 using a No. 325 (45 µm) sieve (see 6.5).

4.5.11.2 Particle size distribution (Classes 2 through 8). The particle size distribution of Classes 2-8 shall be determined in accordance with ASTM B214. All percentages shall be by weight using sieves conforming to ASTM E11 (see 6.5).

4.5.11.3 Particle size distribution (Class 9). The particle size distribution shall be determined in accordance with Method 201.1 of Standard MIL-STD-1234 except the gyration rate shall be 268 – 315 gyrations per minute (see 6.5).

4.5.12 Determination of average particle size (Classes 9 and 10). Determine the average particle size of the aluminum powder in accordance with the method described in ASTM B330 (see 6.5).

4.5.13 Apparent density. The apparent density of classes 1, 4, and 9 material shall be determined in accordance with ASTM B329. For all other classes, the apparent density shall be determined in accordance with ASTM B417.

4.5.14 Process verification. The aluminum type shall be demonstrated to be manufactured using the required process (see 6.4).

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4.5.15 Workmanship. Take approximately 200 to 300 grams of the material and spread the composition out on a clean sheet of white paper. The sample shall be examined with unaided eyes by personnel with normal vision or normal corrected vision. The sample shall be uniform in quality and free from lumps, grit, visible impurities, foreign matter or other defects that would render the material unsuitable for the intended use. Failure to meet the workmanship inspection and/or visual inspection of workmanship during the test methods of 4.5 shall result in lot rejection.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point' packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Aluminum powder covered by this specification is intended for use as listed in Table VII.

Table VII. Intended use.

Type and grade		Intended use
Type I		
Grade A	Class 1	Primer composition
Grade B	Class 2, 2A, 3	Pyrotechnics
Type II		
Grade C	Class 4, 4A	Pyrotechnics
	Class 5, 5A	Plain incendiary thermite
	Class 6	High explosive incendiary projectiles
Type III		
Grade C	Class 6, 7, 8	Heavy explosive
	Class 7	Minol and tritonal loaded items
Grade G	Class 9	Bomb fill, primer compositions, and pyrotechnic compositions
Grade H	Class 10	Rocket motor for the M913

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6.1.1 Historical reference. Historic references to Grades D, E, and F should mean Grade C. Historic references to MIL-DTL-14067 Type II should mean Grade G, and MIL-DTL-14067 Type V should mean Grade H.

6.1.2 Military unique. Aluminum powder covered by this specification is a technical grade aluminum used in military applications such as pyrotechnics, explosives, and primer. The particle size, moisture content, and other chemical characteristics required herein are significant.

6.2 Material safety data sheets. Contracting officers should identify those activities requiring copies of the completed material safety data sheets (MSDS) prepared in accordance with FED-STD-313. The pertinent government mailing addresses for submission of data are listed in FED-STD-313, and 29 CFR 1910.1200 requires that the MSDS for each hazardous chemical used in an operation must identify the activities requiring copies of the MSDS.

6.3 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, grade, and class required (see 1.2).
c. Issue of the DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. Whether a first article sample is required (see 3.1.1).
e. Packaging requirements (see 5.1).

6.4 Definitions.

6.4.1 Flaking. A stamp mill or other process crushes aluminum stock material in a protective atmosphere to produce thin flakes or leaflets.

6.4.2 Graining. A process which reduces aluminum stock material to small particles with a granular appearance.

6.4.3 Atomization. Dispersion of a molten metal into a spray, the spheroidal droplets of which are then allowed to freeze under the primary influence of surface tension.

6.4.4 Batch. That quantity of material that has been manufactured by some unit chemical process or subjected to some physical mixing intended to make the final product substantially uniform.

6.5 Additional particle size requirements and analysis. Programs and suppliers may consider it desirable to impose additional requirements on particle size above and beyond those within the tolerances specified so as to specify further refinement of requirements or receive product optimized for their manufacturing processes. Furthermore, they may consider using alternate or additional methods and equipment to specify, characterize, and analyze size. Such analysis techniques may include:

a. Gravitational sedimentation (see ASTM B761 and ISO 10076)

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b. Microscopy (see ISO 13322 and BS 3406-4)
c. Laser diffraction (see ASTM B822 and ISO 13320)

Those who are interested in alternate characterization techniques are cautioned that analysis and results from one technique are generally not directly comparable to other techniques due to numerous factors including particle morphology, measurement technique, etc. Beneficial information may be found on alternate techniques from the National Institute of Standards and Technology (NIST) Special Publication 960-1, Recommended Practice Guide on Particle Size Characterization, available at www.nist.gov.

6.6 Diethyl ether. Diethyl ether used for fatty/oily matter extraction should be stabilized by butylated hydroxytoluene (BHT).

6.7 Silicon test method. The following notes are included for informational purposes regarding the silicon test method in 4.5.5. This test method originally appeared in ASTM E34.

6.7.1 Achieving complete dissolution. With certain alloys it may be necessary to evaporate the caustic solution to pastiness in order to completely dissolve the silicon. Care should be taken to avoid mechanical loss by spattering.

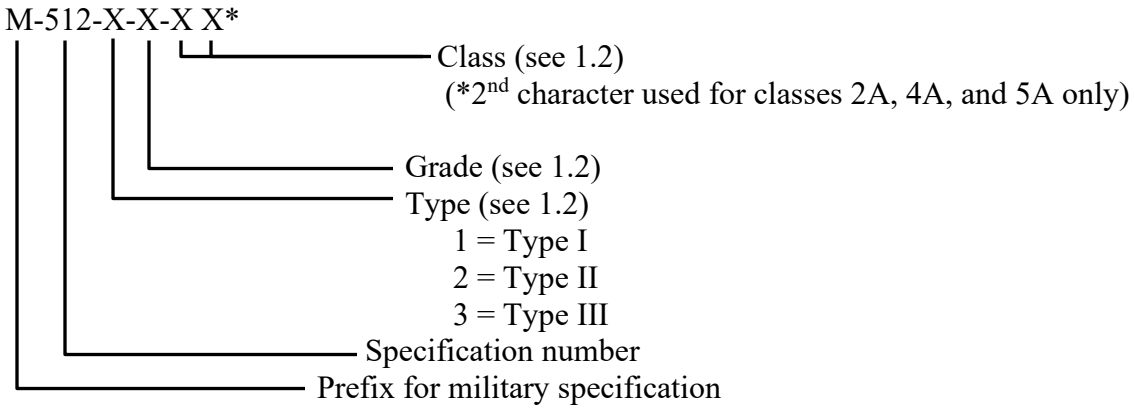
6.7.2 Solution cloudiness and excessive heating. Cloudiness (gray) of the solution at this point usually indicates incomplete dissolution of the silicon and another portion of the sample must be taken. Prolonging the heating after the NaOH attack will probably correct the difficulty. However, excessively prolonged heating will tend to cause precipitation of the manganese.

6.7.3 Brownish-yellow solutions. If the sample contains high manganese, the solution may be colored brownish-yellow due to suspended manganese dioxide (MnO_2). To bleach the color, add a saturated sodium sulfite (Na_2SO_3) solution drop-wise to the hot solution. Avoid any appreciable excess. Add potassium permanganate ($KMnO_4$) solution (3.2 g/L) until the solution is tinted pink. Finally add just enough oxalic acid (10 g/L) to destroy the pink color. Transfer the solution to a 100-mL volumetric flask, dilute to volume, and mix.

6.7.4 Correction for reference solution. For routine work, the reference solution correction may be omitted unless the test solution is colored yellow.

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6.8 Part or identifying number (PIN). The PIN to be used for aluminum powder acquired to this specification is created as follows:



6.9 Type III Grades G and H. Type III Grades G and H material were previously in MIL-DTL-14067C. They were transferred to this specification revision D by ECP R17Q2013.

6.10 Changes incorporated. This specification was updated from MIL-DTL-512C to MIL-DTL-512D with Amendment 1 as part of the Munitions Chemical Specification Modernization (MCSM) program. Types of material were moved added (see 6.9). All test procedures were verified and updated as needed to produce valid results. Reference documents have been verified, updated, and hyperlinked to sources. Details of these changes are documented in the ECPs and ERR listed below.

Specification ID	ECP / ERR
MIL-DTL-512D	R17Q2013
MIL-DTL-512D w/ Amendment 1	Y24Q3005

6.11 Subject term (key word) listing.

Explosives
Incendiaries
Metal
Primer
Propellants
Pyrotechnics
Munitions Chemical Specification Modernization (MCSM)

6.12 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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CONCLUDING MATERIAL

Custodian:
Army – AR
DLA – GS

Reviewer:
Army – EA

Preparing activity:
Army – AR
(Project 6810-2025-008)

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